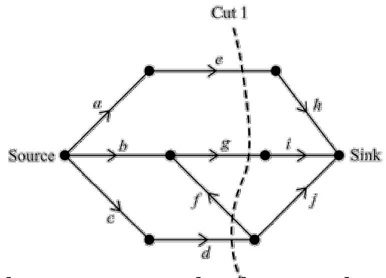
Critical Path Analysis Topic Test 1 Solution

Total Marks: 30

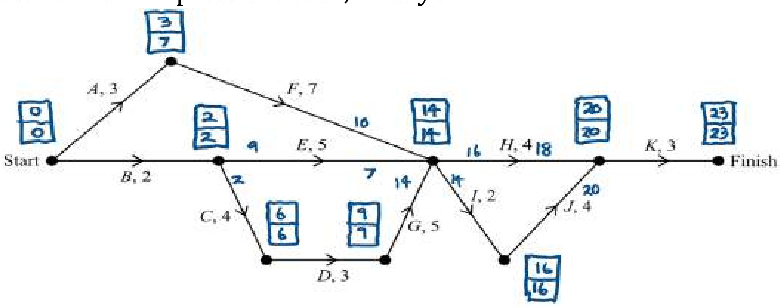
Time: 50 minutes

1 Water pipes connect 8 locations. The flow network below shows the capacities of the pipes. What is the capacity of cut 1? 1



edge f is ignored because it flows from sink to source

2 The network diagram shows tasks that must be completed to finish a project. Each edge weight represents the time taken to complete the task, in days.



a. What is the earliest start time (EST) of activity H? 2

14 days

b. What is minimum completion time for the project? 1

23 days

c. What is the critical path for the project? 1

B-C-D-G-I-J-K

d. What is the float time of activity E? 1

float time = LST_{next} − EST − duration

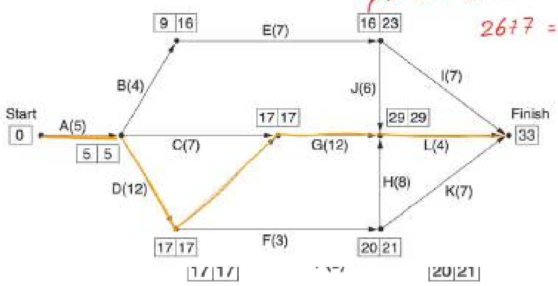
= 14 − 2 − 5

= 7 days

3 The network below shows the earliest start and latest start times of each activity for a project. All times are in terms of hours. If activity I is delayed by 10 hours from its earliest starting time, 16 hours, how many hours will the project finish be delayed?

+10 so start at 26

26+7 = 33



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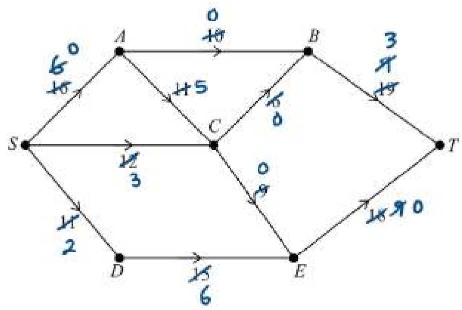
4 The flow of water through a series of pipes is shown in the network below. The water flows from the source (S) to the sink (T) and the numbers on the edges show the maximum capacity of each pipe in litres per minute.

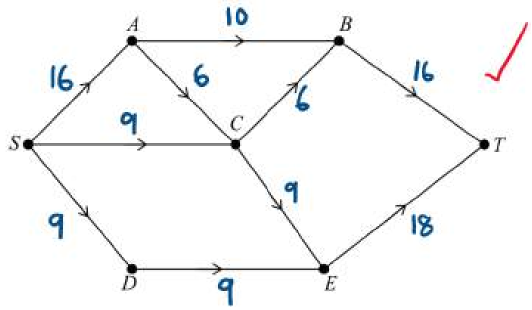


SABT 10
SACBT 6
SCET 9
SDET 9
34

a. Find the maximum flow from S to T. 2

maximum flow = 10 + 6 + 9 + 9
= 34 L/min

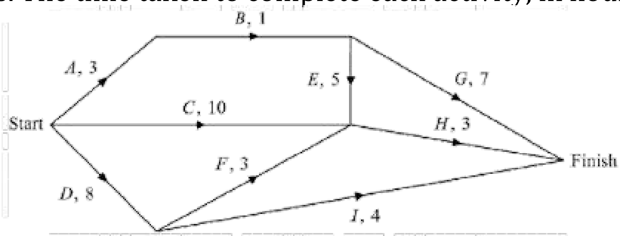
b. On the diagram below, label the flows along each edge that would be required to achieve the maximum flow. 1



c. By increasing the capacity of one edge in the network, the maximum flow could be increased by 3 litres per minute. Identify the edge that could be increased. 1

The minimum cut passes through AB, CB and ET so only changes to these edges will affect the minimum flow. Only edge CB has the potential for increase by 3

5 The directed network below shows the sequence of activities A to I that is required to complete a manufacturing process. The time taken to complete each activity, in hours, is also shown.



a. Determine the minimum completion time for the direct network and state the critical path. 2

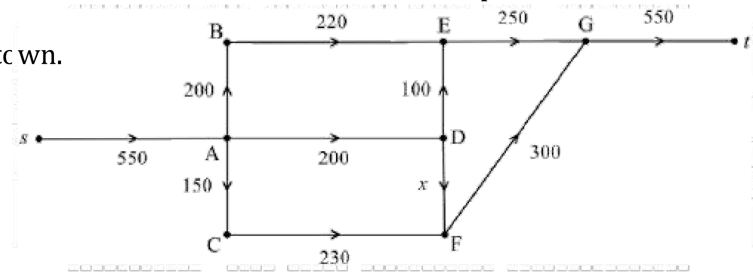
Critical path: DFH
Minimum completion time: 14 hours.

b. What is the float time of activity G? Float time $= 14 - 4 - 7 = 3$ hours 1

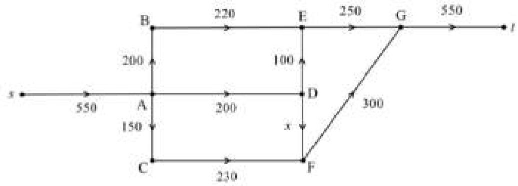
c. What is the latest starting time for activity A? 1

LST(A) is after 2 hours

6 The network diagram shows the maximum flow of traffic, in vehicles per hour, between along several streets in a congested town.



a. It is known that the minimum cut of this network cuts through edge DF. The maximum flow of the network is 530 vehicle per hour. Determine the value of x. 2

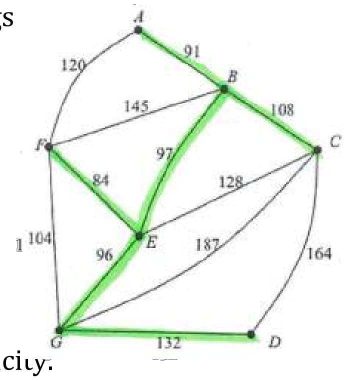


$x + 150 + 250 = 530$
$x = 530 - 400$
So $x = 130$

b. A new road from D to G is being built. The flow capacity of this road is 30 vehicles per hour. Will this road increase the maximum flow of the entire network to 560 vehicles per hour? Justify your answer. 1

No, the capacity from $s \to A$ is still 550 vehicles per hour, so the max flow cannot be more than 550.

7 The network diagram shows the electrical wires connecting 7 buildings on a school campus. The vertices A to G represent the buildings. The weights on the edges represent the length of wire in metres used to connect the buildings.



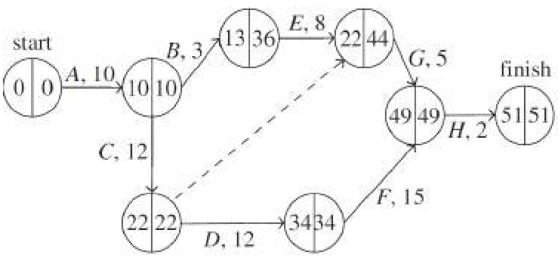
= 608m

a. Draw the minimum spanning tree to show the minimum length of wire required to connect all the buildings with electricity. 2

b. If the electrician charges \$55 per hour and the wire costs \$26.50 per metre, how much will it cost to wire the school if they are working on the job from 7:30am – 4pm? 1

$(608 * 26.50) + (55 \times 8.5) = \$16\,579.50$

8 The network diagram shows the activities necessary to complete a project. Activity duration is in hours.



a. On the network diagram, write the earliest starting times (ESTs) and the latest starting times (LSTs). 2

b. What is the critical path? 1

The critical path is $A \to C \to D \to F \to H$.

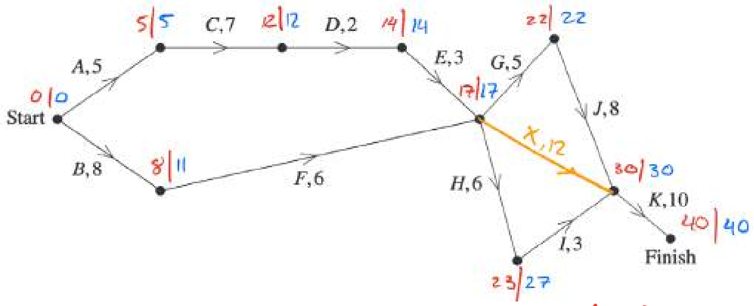
c. Determine the float time of activity G. 1

float time of activity G = LST − EST = 44 − 22 = 22 hours

Alternatively, since activity G has other paths leading to the next activity:

float time of activity $G = \text{LST}_{\text{next}} - \text{EST} - \text{activity time}$
$= 49 - 22 - 5 = 22$ hours

9 A project requires completion of 11 tasks A, B, C, . . . , K. A network diagram for the project giving the completion time for each task, in minutes, is shown.



a. Find the minimum time to complete the project. 40 1

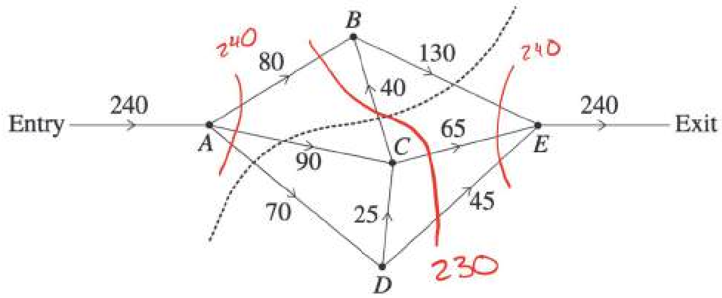
b. State the critical path for this project. 1

A C D E G J K

c. A new task, X, is to be added to the project. The earliest starting time for X is 17 minutes, the latest starting time for X is 18 minutes and X has a completion time of 12 minutes. Add task X to the given network diagram above AND state the float time for this task. 2

1 minute

10 A museum is planning an exhibition using five rooms. The vertices A, B, C, D and E represent the five rooms. The nu mbers on the edges represent the maximum number of people per hour who can pass through the security checkpoints between the rooms.



a. What is the capacity of the cut shown? 1

130 + 90 + 70 = 290

b. What is the flow capacity of the network? 230 1

Critical Path Analysis Topic Test 2 Solution

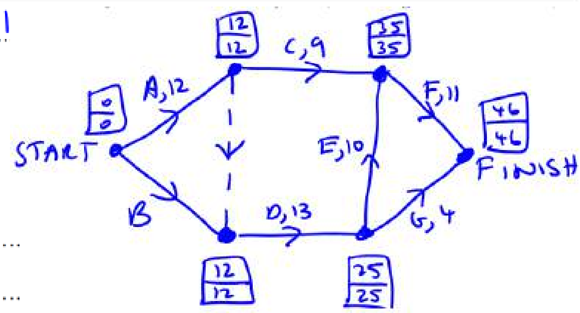
Total Marks: 30

Time: 50 minutes

1 Consider the following activity table. The critical path is A, D, E, F. The critical time for the project is 46 days.

Activity	Duration (days)	Prerequisites
A	12	None
B	?	None
C	?	A
D	13	A, B
E	10	D
F	?	C, E
G	?	D

a. Draw an activity chart for this activity table, with weights for activities A, D and E only. 2



b. Find the weight of activity F. 46 − 12 − 1 1

c. The float time of activity G is 17 days. 1

Find the weight of activity G.

25 + 17 = 42

46 − 42 = 4

∴ Weight of 4 days

d. The float time of activity C is 14 days. Find the weight of activity C. 1

12 + 14 = 26

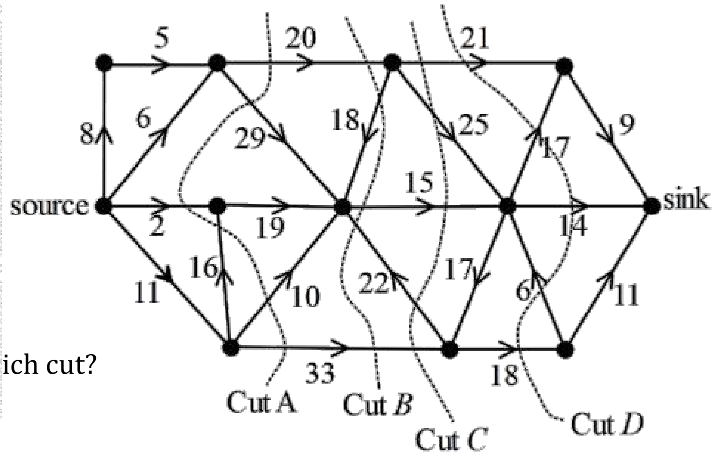
35 − 26 = 9

∴ Weight of 9 days

e. What is the longest possible duration of activity B? 12 days 1

2 In the network below, the values on the edges give the maximum flow possible between each pair of vertices. The arrows show the direction of flow. 1

The maximum flow between source and sink through this network is given by which cut?

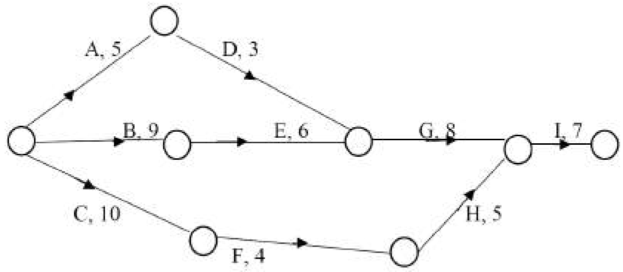


Cut B

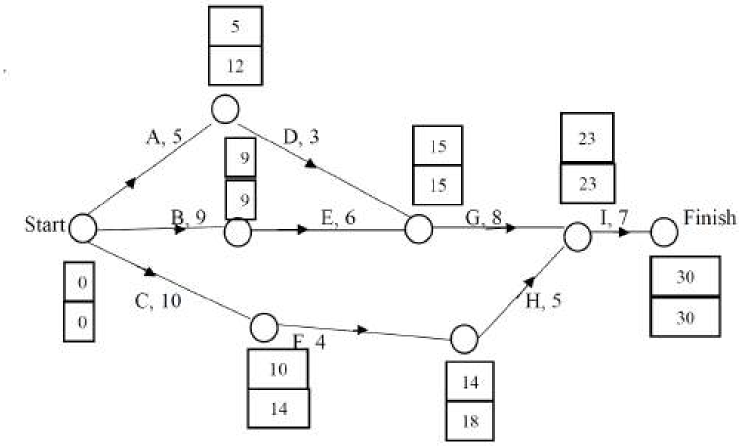
3 Five musicians are to record an album that involves nine activities. The activities and their immediate predecessors are shown in the table below. The duration of each activity is shown in hours.

Activity	Duration	Predecessor
A	5	-
B	9	-
C	10	-
D	3	A
E	6	B
F	4	C
G	8	D, E
H	5	F
I	7	G, H

a. Use the information in the above table to complete the network below by including activities G, H and I. 1

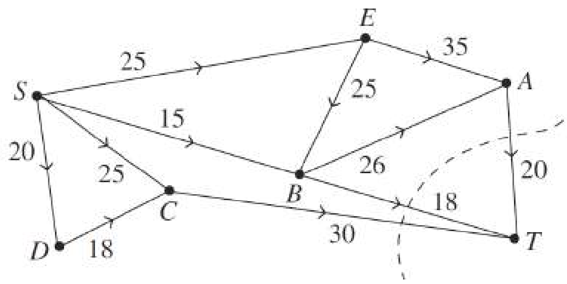


b. Determine the critical path and the minimum completion time for this project. 2



BEGI is the critical path since the activities where EST (earliest start time) = LST (latest start time). The minimum completion time is 30 hours.

4 In the directed network, the source is represented by S and the sink is represented by T.



a. Find the capacity of the minimum cut shown above. $30 + 18 + 20 = 68$ 1

b. Show that the maximum flow is equal to the minimum cut. 2

c. What is the outflow at B? 1

possible outflow at $B = 26 + 18 = 44$
inflow at $B = 15 + 25 = 40$
Therefore, the outflow at B is 40 (the maximum outflow of the vertex is the smaller of the inflow or outflow of the vertex).

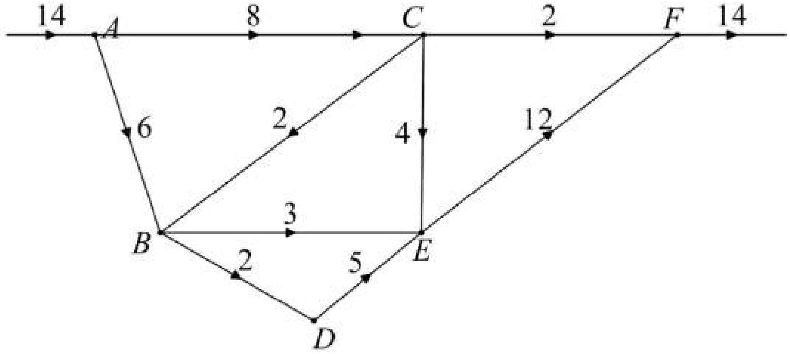
The flows that give a maximum are as follows.

- *SEAT*: 20
- *SEBT*: 5 (Since *SE* is 25 – 20 = 5 from path *SEAT*.)
- *SBT*: 13 (Since *BT* is 18 – 5 = 13 from path *SEBT*.)
- *SCT*: 25
- *SDCT*: 5 (Since *CT* is 30 – 25 = 5 from path *SCT*)

$20 + 5 + 13 + 25 + 5 = 68$

Therefore, the maximum flow is equal to the minimum cut.

5 The network diagram shows a series of footbridges and treetop walks in a National Park. The vertices A, B, C, D, E and F represent six platforms. The edges represent the footbridges that connect the platforms. The numbers on the edges indicate the maximum capacity of people allowed on a footbridge.

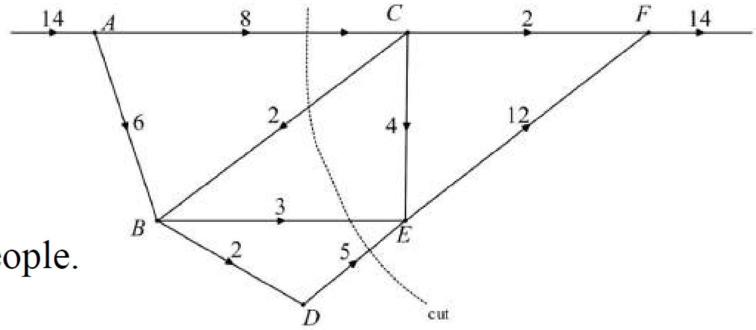


a. Determine the maximum flow of the network. 2

The minimum cut passes through BD, BE, CE and CF.

$\therefore$ Maximum flow $= 2 + 3 + 4 + 2 = 11$ people.

b. A cut is added to the network as shown. What is the capacity of this cut? 2

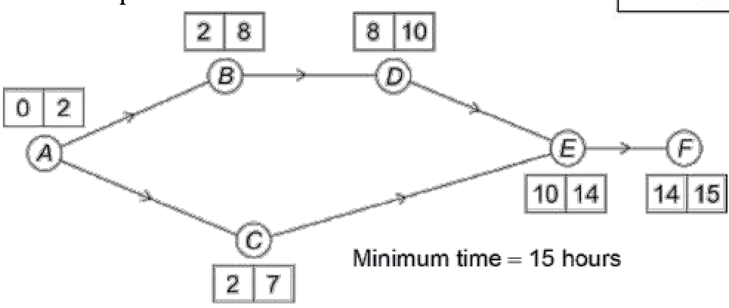


$5 + 3 + 8 = 16$ people.

6 A project requires activities A to F to be completed. The activity chart shows the immediate prerequisite(s) and duration for each activity.

Activity	*Immediate prerequisite(s)*	*Duration in hours*
A	—	2
B	A	6
C	A	5
D	B	2
E	C, D	4
F	E	1

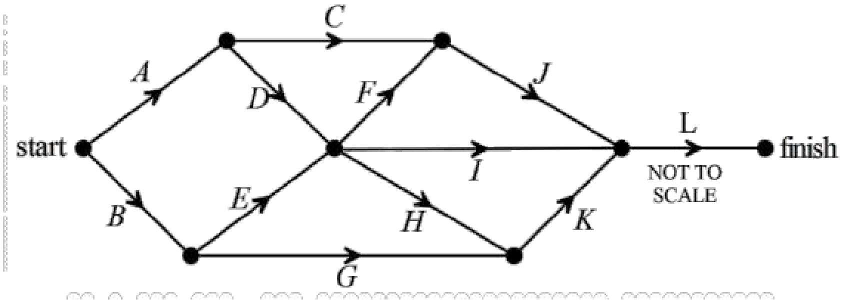
a. By drawing a network diagram, determine the minimum time for the project to be completed. 2



b. Determine the float time of the non-critical activity. 1

The float time for the non-critical activity (C) is 3 hours.

7 The network below shows the activities that are needed to finish a particular project.



a. What is the total number of activities that need to be completed before Activity K may begin? 1

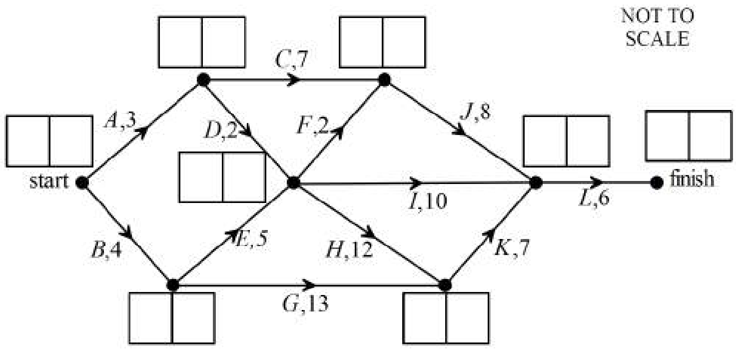
6

b. The duration of every activity is initially 2 hours. For an extra cost, the completion times of both Activity C and Activity G can be halved. If this occurs, what effect will it have on the completion time of the project? Explain your answer. 2

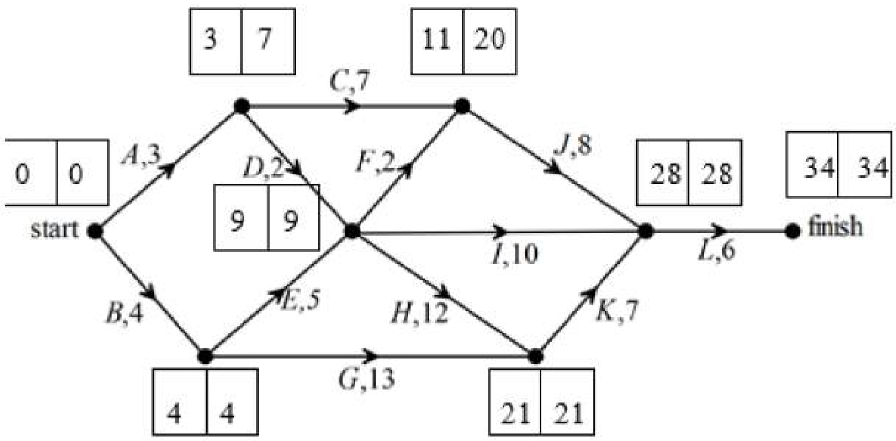
Since all activities have the same duration, there are three critical paths: *BEHKL*, *ADHKL* and *ADFJL*. Activities *C* and *G* are not on the critical paths so halving their times will have no effect on the completion time on the project.

c. Before the commencement of the project, the company undertaking it had to cut back on some of its workers. This affected the completion time of each activity. They were given a deadline of one-and-a-half days. 3

The duration of each activity is shown in the network below.



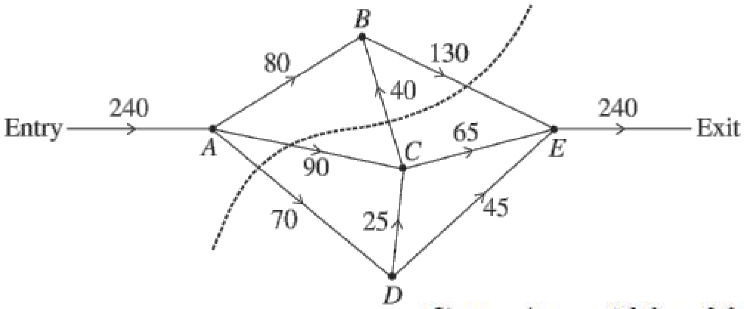
Complete the diagram to find the earliest starting times (EST) and latest start times (LST) for all activities and hence, calculate the float time for Activity C.



1.5 days = 36 hours

∴Float time for Activity *C* is 20 − 11 + 2 = 11 hours

8 A museum is planning an exhibition using five rooms.The museum manager draws a network to help plan the exhibition. The vertices A, B, C, D and E represent the five rooms. The numbers on the edges represent the maximum number of people per hour who can pass through the security checkpoints between the rooms.



a. What is the capacity of the cut shown? $\text{Capacity} = 130 + 90 + 70 = 290$ 1

b. The museum manager is planning for a maximum of 240 visitors to pass through the exhibition each hour. By using the 'minimum cut – maximum flow' theorem, the manager determines that the plan does not provide sufficient flow capacity. 2

Draw the minimum cut onto the network below and recommend a change that the manager could make to one or more security checkpoints to increase the flow capacity to 240 visitors per hour.

Recommended change: Increase the capacity of checkpoint between C and B from 40 to 50.